CSE 516 Major Project — M3 Results and Analysis

Abstract

This report evaluates a temporal Bayesian skill-estimation PGM and an extension that incorporates individual-level performance observations. The baseline uses team match outcomes only; the extension uses the same outcomes plus synthetic individual observations. Experiments use synthetic doubles data with known latent player skills. The principal result is lower individual skill-estimation error for the extension across ten random seeds. The paired comparison reports a mean MAE reduction of 0.1740, paired t-test p-value 0.00546552 and bootstrap 95% confidence interval [0.0971, 0.2736]. The findings are limited to the controlled synthetic data-generating assumptions.

1. Objective and Scope

M3 focuses on evaluating the proposed extension to the temporal Bayesian skill-estimation model. The evaluation compares a team-outcome-only baseline with an extended model incorporating individual-level performance observations. The experiments examine estimation accuracy, predictive performance, statistical significance, robustness, uncertainty and computational behaviour. The objective is to assess the effectiveness and limitations of the proposed approach using controlled synthetic data.

2. Limitation and Proposed Method

The baseline model uses team match outcomes to estimate player skills. The extension incorporates additional individual-level performance observations to improve the estimation of individual abilities. Since the supplied ATP history dataset does not contain the required individual statistics, synthetic observations are used for controlled experimentation. Both models are evaluated using the same generated matches and random seeds.

3. Model and Implementation

The temporal model represents each player's skill as a latent variable evolving through a Gaussian random walk. Team outcomes follow a logistic likelihood based on the difference between team-average skills. In the extension, each individual performance observation has a Gaussian likelihood centred on that player's latent skill. MAP inference is used, with a Laplace approximation for posterior uncertainty. This is a simplified temporal Bayesian PGM, not a reproduction of the complete published TTT EP implementation.

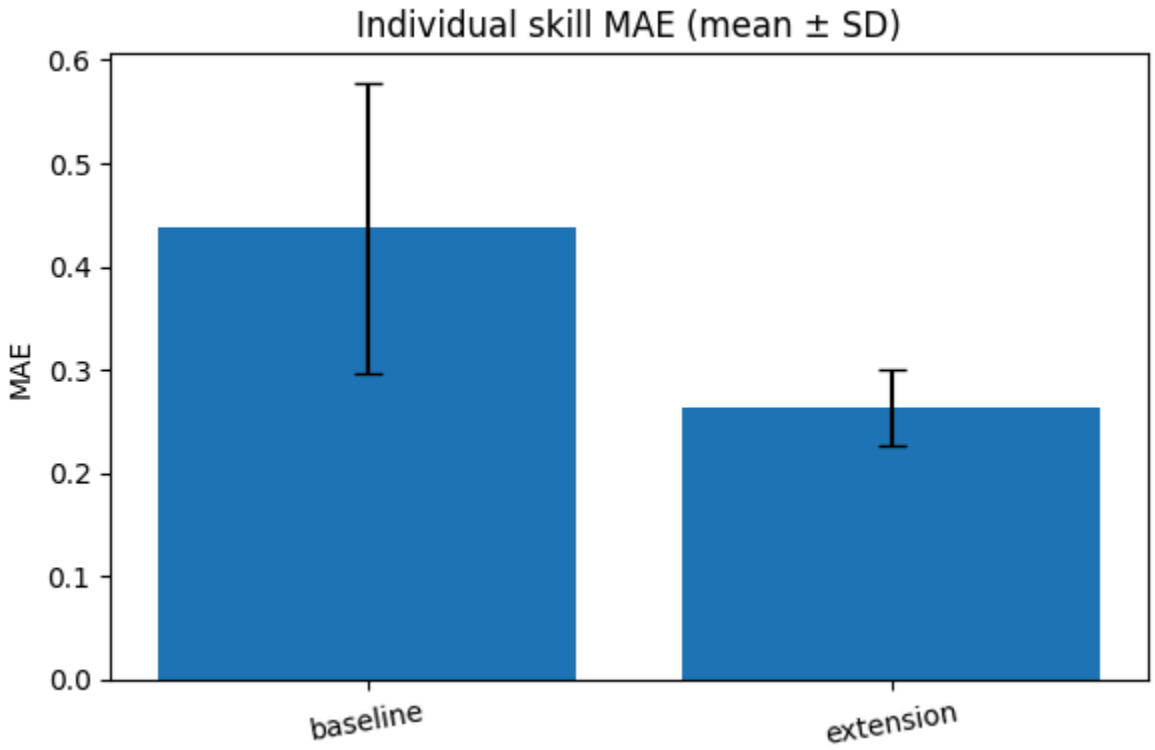
4. Experimental Protocol

The main experiment uses ten random seeds (42–51) and a chronological 70:30 time split. For each seed, both models use the same generated data. The primary KPIs are individual skill MAE and RMSE against known simulated ground truth. Secondary KPIs are held-out team-outcome log-loss and Brier score. Runtime, optimisation iterations, optimiser success and approximate posterior standard deviation are also recorded. Paired tests compare the models seed by seed.

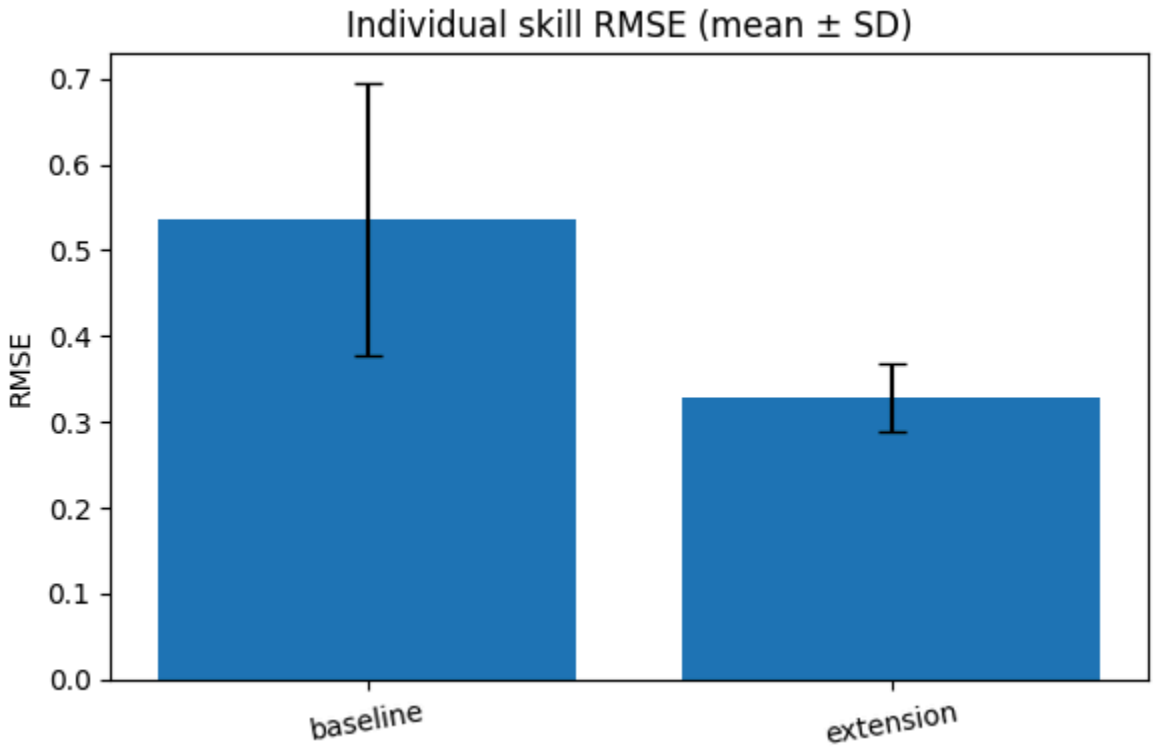
5. Main Results

The ten-seed results are:

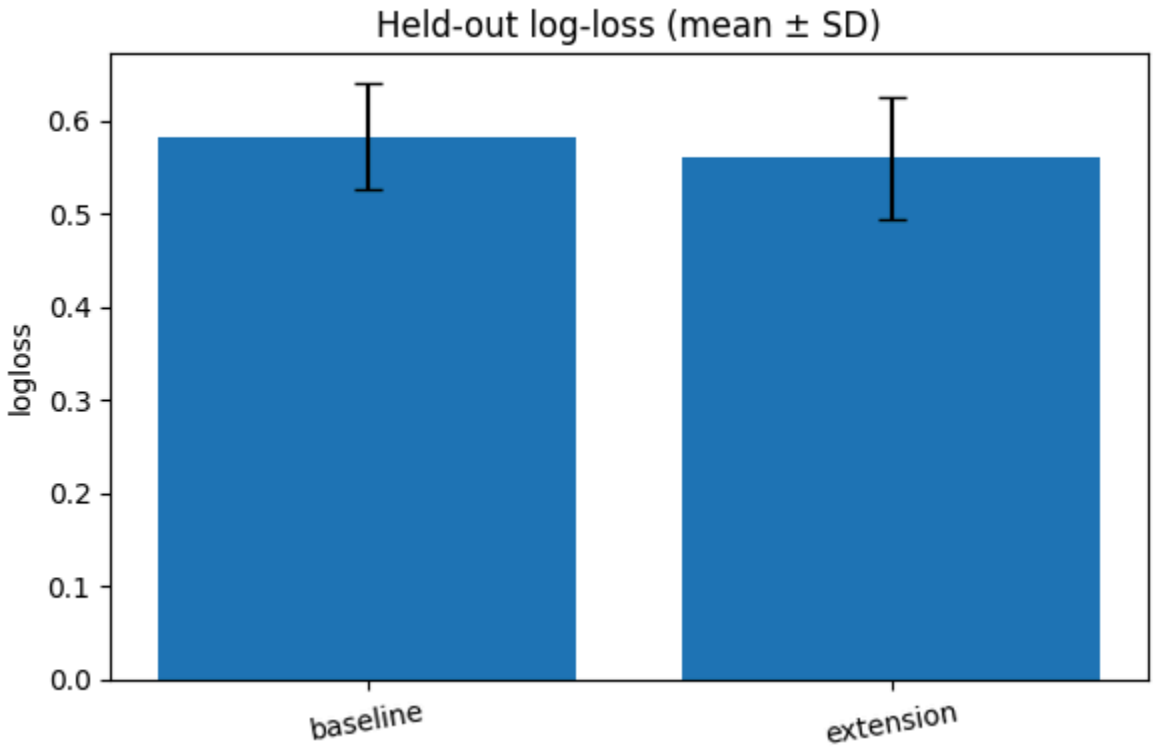
Baseline MAE 0.4372 ± 0.1400 and extension MAE 0.2632 ± 0.0366



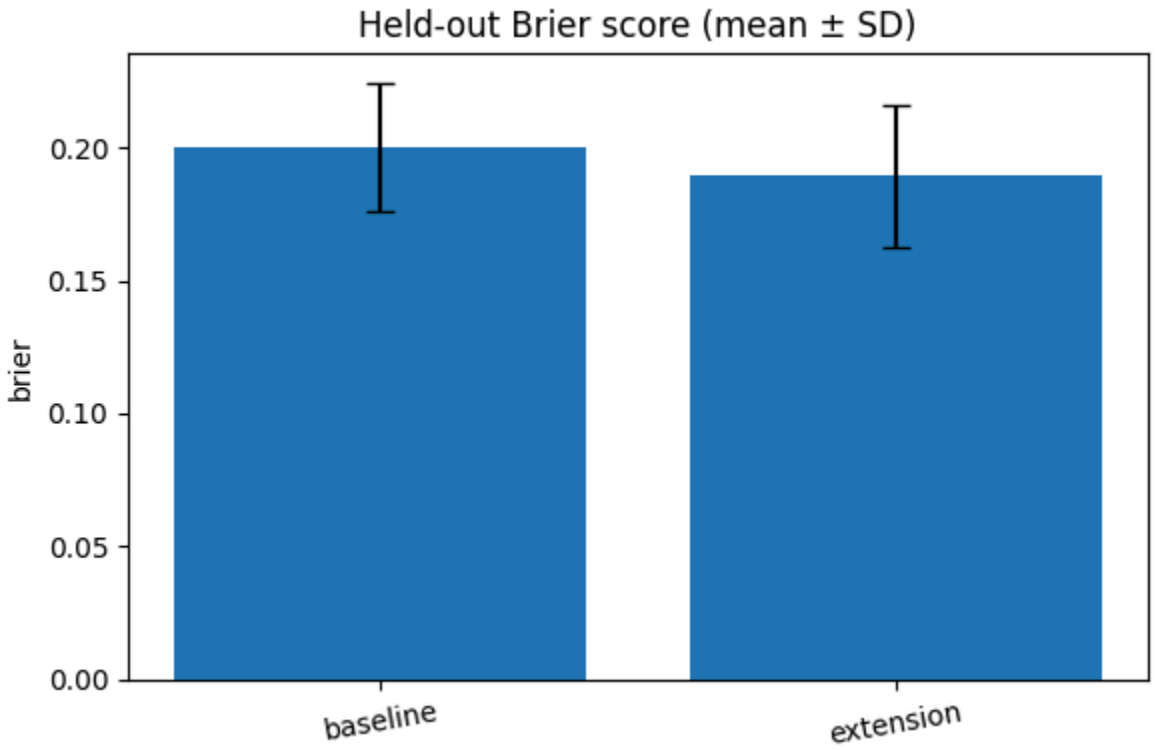
Baseline RMSE 0.5357 ± 0.1586 and extension RMSE 0.3284 ± 0.0395.



Baseline log-loss is 0.5835 ± 0.0568 and extension log-loss is 0.5601 ± 0.0655.



Baseline Brier score is 0.1999 ± 0.0242 and extension Brier score is 0.1893 ± 0.0269.



Mean runtime is 1.4739 ± 0.4884 seconds for the baseline and 1.1377 ± 0.3502 seconds for the extension in the recorded environment. Runtime is hardware- and session-dependent.

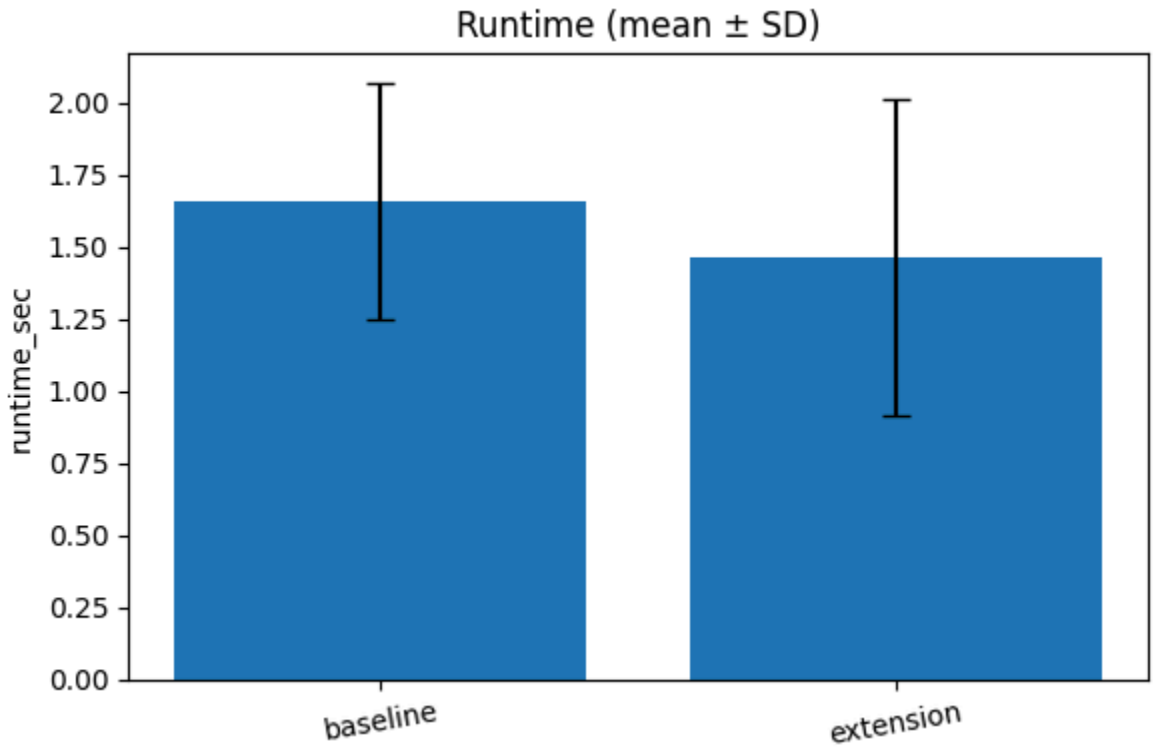


Table 1: Main KPI Summary

Model	MAE (Mean ± SD)	RMSE (Mean ± SD)	Log-loss (Mean ± SD)	Brier Score (Mean ± SD)
Baseline	0.4372 ± 0.1400	0.5357 ± 0.1586	0.5835 ± 0.0568	0.1999 ± 0.0242
Extension	0.2632 ± 0.0366	0.3284 ± 0.0395	0.5601 ± 0.0655	0.1893 ± 0.0269

6. Statistical Analysis

The paired MAE difference is defined as baseline MAE minus extension MAE, so a positive value favours the extension. The mean paired reduction is 0.1740. The paired t-test gives p = 0.00546552. The bootstrap 95%

confidence interval is [0.0971, 0.2736]. Since this interval is above zero and the p-value is below 0.05, the results provide evidence of lower MAE for the extension across the ten tested seeds. The statistical conclusion applies to this synthetic experimental setup and should not be generalised to real tennis data without further validation.

7. Robustness Analysis

The robustness experiments examine three dimensions: number of matches per time point, missing individual observations, and noise in individual observations. The extension achieves lower MAE than the baseline at all three tested sample sizes. Its MAE increases as observations are removed and as observation noise increases, but it remains below the baseline at the highest tested missing-observation and noise levels. Graph-density robustness was excluded because match connectivity is an implicit chronological artifact of the random-walk formulation, making it difficult to parameterise independently.

Table 2: Sample Size Robustness (Matches per time)

Matches	Model	MAE	Log-loss	Brier	Runtime (sec)
6	Baseline	0.5037	0.6019	0.2047	1.0311
6	Extension	0.2989	0.5815	0.1966	0.8745
12	Baseline	0.4372	0.5835	0.1999	0.8825
12	Extension	0.2632	0.5601	0.1893	0.7376
24	Baseline	0.4527	0.5788	0.1978	0.9531
24	Extension	0.2392	0.5580	0.1891	0.9064

Table 3: Missing Observation Robustness

Missing Rate	Model	MAE	Log-loss	Brier
0.00	Baseline	0.4372	0.5835	0.1999
0.00	Extension	0.2632	0.5601	0.1893
0.25	Baseline	0.4372	0.5835	0.1999
0.25	Extension	0.2656	0.5604	0.1895
0.50	Baseline	0.4372	0.5835	0.1999
0.50	Extension	0.2740	0.5609	0.1896
0.75	Baseline	0.4372	0.5835	0.1999
0.75	Extension	0.2937	0.5612	0.1895

Table 4: Observation Noise Robustness

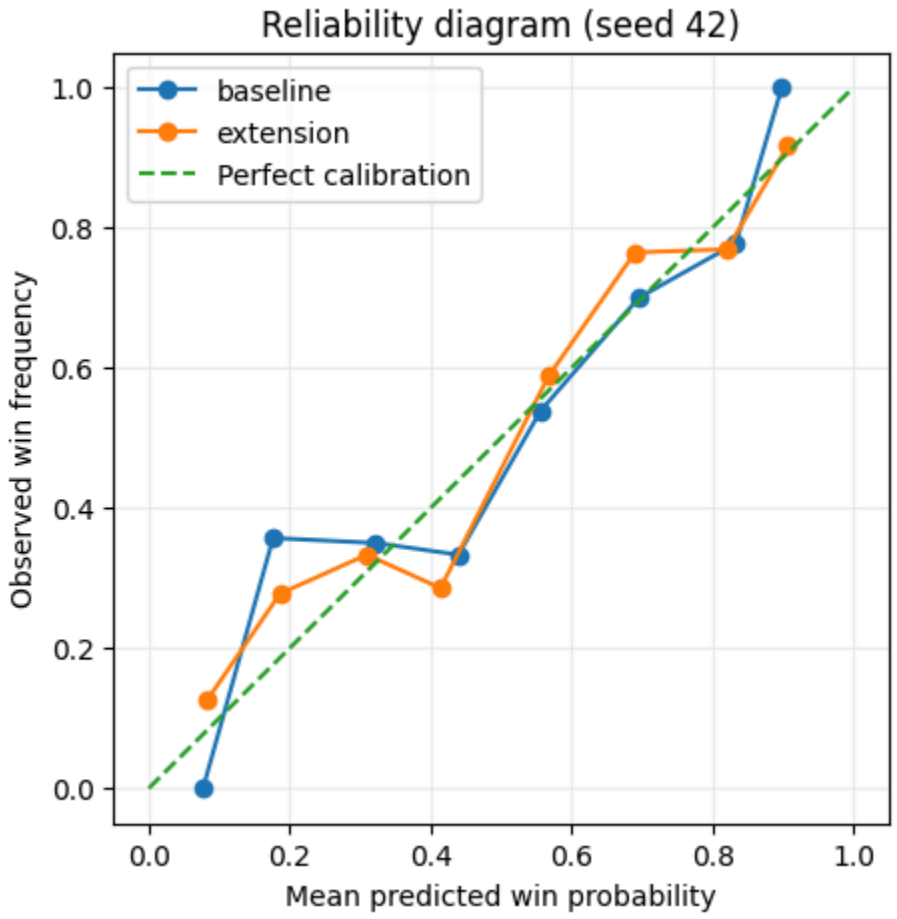
Noise SD	Model	MAE	Log-loss	Brier
0.25	Baseline	0.4372	0.5835	0.1999
0.25	Extension	0.2358	0.5581	0.1884
0.65	Baseline	0.4372	0.5835	0.1999
0.65	Extension	0.2632	0.5601	0.1893
1.00	Baseline	0.4372	0.5835	0.1999
1.00	Extension	0.2822	0.5619	0.1901

Noise SD	Model	MAE	Log-loss	Brier
1.50	Baseline	0.4372	0.5835	0.1999
1.50	Extension	0.3028	0.5642	0.1912

8. Probability Quality and Calibration

Held-out log-loss and Brier score are lower on average for the extension in the ten-seed experiment. A reliability diagram was generated for representative seed 42.

The reliability diagram provides a representative visual assessment of predicted probabilities for seed 42. It is not an aggregate calibration assessment across all ten random seeds.



9. Runtime, Convergence and Uncertainty

The notebook records runtime, optimisation iterations, optimiser success and mean approximate posterior standard deviation. The runtime comparison is specific to the execution environment. The posterior standard deviation is based on a Laplace approximation and should be interpreted as approximate model uncertainty, not a guaranteed calibrated interval.

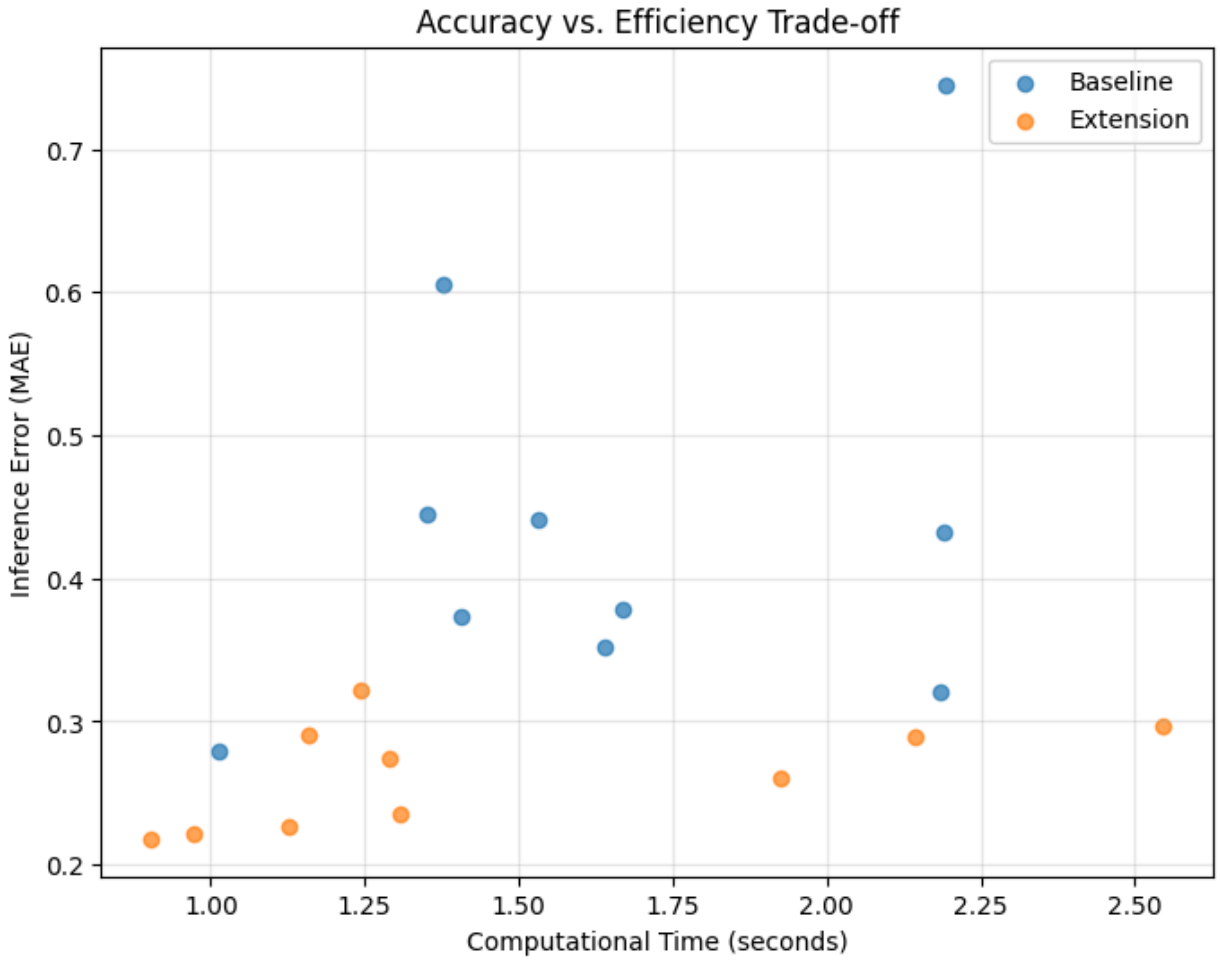


Table 5: Runtime and Convergence Analysis

Model	Mean Runtime (s)	SD Runtime (s)	Mean Iterations	Optimizer Success Rate	Mean Posterior SD
Baseline	1.4739	0.4884	164.5	1.0	0.6182

Model	Mean Runtime (s)	SD Runtime (s)	Mean Iterations	Optimizer Success Rate	Mean Posterior SD
Extension	1.1377	0.3502	67.4	1.0	0.3182

Both models achieved a 100% optimiser success rate in the recorded experiments. The extension required fewer mean optimisation iterations in this implementation. Its lower recorded runtime should be interpreted cautiously because runtime depends on the execution environment. The lower approximate posterior standard deviation indicates lower estimated uncertainty, although the Laplace approximation does not guarantee calibrated uncertainty intervals.

10. Ablation and Player-Level Error Analysis

The baseline-versus-extension comparison functions as an ablation of the individual-observation likelihood. The baseline excludes individual-level performance observations, whereas the extension incorporates them. The player-wise error table for seed 42 shows that the extension does not improve every individual player. For example, Player 0's MAE increases from 0.2174 to 0.4651, whereas Player 2's MAE decreases from 0.9457 to 0.1742. This demonstrates that aggregate improvement does not guarantee improvement for every player. The current analysis does not establish the specific reason for the increase in Player 0's error.

Table 6: Player-Level Error Analysis (Seed 42)

Player	Baseline MAE	Extension MAE
0	0.2174	0.4651
1	0.1381	0.1766
2	0.9457	0.1742
3	0.2884	0.2000

Player	Baseline MAE	Extension MAE
4	0.1860	0.1901
5	0.7783	0.4711
6	0.2836	0.1636
7	0.5621	0.1187
8	0.1451	0.1195
9	0.3954	0.1393
10	0.2317	0.3379
11	0.3003	0.0601

11. Interpretation

The reduction in MAE and RMSE indicates that individual-level observations help recover simulated player skills more accurately than team outcomes alone. The improvement is supported by the paired statistical analysis across ten seeds. However, the smaller improvements in log-loss and Brier score suggest that better individual skill estimation does not necessarily translate into equally large improvements in predicting team outcomes. Robustness tests further show that the benefit decreases when observations become less available or noisier. Player-wise analysis also reveals that the extension does not improve every individual player. These findings support the proposed approach under the synthetic experimental assumptions, while highlighting the need for validation using real player-level statistics.

12. Revised SOTA Positioning

The project should be positioned as a simplified temporal Bayesian skill-estimation prototype inspired by TrueSkill Through Time, extended with individual-level auxiliary observations. It is not a full implementation or benchmark

victory over the state of the art. The evidence demonstrates a controlled proof-of-concept on synthetic doubles data. A stronger next step would be evaluation with real individual-level performance statistics and a more realistic observation likelihood.

13. Limitations and Future Work

The ATP dataset used in the project does not contain the individual statistics needed to validate the extension on real matches. The synthetic observation is generated from latent skill plus Gaussian noise, which makes it informative by construction. The current model uses MAP and Laplace approximation and does not reproduce the complete TTT EP algorithm. Future work should acquire suitable player-level match statistics, model their dependence on opponents and match context, test sensitivity to observation-model misspecification, and evaluate on held-out real matches.

14. Conclusion

The M3 experiments provide evidence that adding individual-level synthetic observations improves aggregate individual-skill recovery compared with using team results alone in the tested temporal Bayesian model. The MAE improvement is supported by a paired test and bootstrap confidence interval across ten seeds. The results also expose limits: noisy or missing observations reduce the benefit, and some individual players are estimated less accurately. The findings are a synthetic proof-of-concept and should be reported with that qualification.